

GradeLocal is an AI feedback assistant I designed and built, demonstrated here with synthetic student data. It drafts personalized feedback emails and retake review sheets from student work. Two design requirements shaped it: every model runs on hardware the teacher owns, so student data never reaches a cloud AI, and the teacher approves every word before a student sees it.

143

passing automated tests

117

commits since April 2026, built solo

1 min

demo video, synthetic data only

THE PROBLEM

Specific, encouraging written feedback helps students most, and it takes teachers the most time. Cloud AI tools would make it faster, but sending student work to a third-party server conflicts with student-data privacy policy. Districts have the same concern: they want the time savings, and they also need a clear answer about where student data goes. **Designed for:** the teacher, who reviews and approves; students, who read the feedback; and parents, who may read it with them.

KEY DECISIONS: TRADING AUTOMATION FOR TRUST

Decision	Tradeoff accepted	Why
Every model runs on local hardware; no cloud API sees student work	Smaller, slower models	It answers the privacy question completely, not with a policy promise
The teacher approves every email and sheet before a student sees it	It's never "fully automatic"	Approval is both the compliance mechanism and the quality bar
Math is checked symbolically; a failed step is dropped, never "repaired"	Some questions get no worked solution	A wrong worked solution does more harm than an empty box
When the model can't verify an answer, it's marked "unverified" or left for the teacher	More manual work on hard cases	The tool is honest about what it doesn't know

PROBLEMS FOUND AND FIXED

Problem	Fix
A retake score alone doesn't show what was missed	Built review sheets: quiz screenshots in, a "how to solve it" sheet out, which the teacher edits and approves
The model confidently "solved" a graph question it had misread	Questions with a figure go to the teacher; the model's own guesses are labeled "unverified"
One quiz layout made the questions unreadable	The reader uses row colors; the quiz's own answer line cross-checks but never sets the answer
Explaining a "mistake" on a question left blank doesn't help	Blank questions are taught from scratch instead
Model-written math showed up as raw LaTeX in emails	Math is written as plain text

TESTING

- **143 automated tests pass** on the review-sheet pipeline: reading marks, checking the math, writing steps, rendering, and the server routes.
- **The one-minute demo** uses only synthetic data.

HOW I WORK AS A PM

- **I start from the constraint.** Privacy shaped the product before any features did.
- **I treat trust as a feature.** The tool says "unverified" rather than guess.
- **I design for what users feel:** the app logs time to approve and whether a draft was edited, not model scores.
- **I write decisions down,** with dated records of what was chosen, what was rejected, and why.

Next: ship practice problems only with machine-checked answers. Built solo: 117 commits since April 2026, 143 automated tests. It runs on HKOS, a self-hosted AI stack I built, which I'm glad to walk through in an interview.